R Code

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# ============================================================
# LAB 6: MULTIVARIATE MULTIPLE REGRESSION ANALYSIS
# ============================================================

# OBJECTIVE:
# To fit a multivariate multiple regression model in R
# when there are two response variables and several predictors.
#
# Response variables:
# Y1 = Heating Load
# Y2 = Cooling Load
#
# Predictor variables:
# X1 = Relative Compactness
# X2 = Surface Area
# X3 = Wall Area
# X4 = Roof Area
# X5 = Overall Height
# X6 = Orientation
# X7 = Glazing Area
#
# ============================================================
# 1. LOAD THE REQUIRED PACKAGE
# ============================================================

if (!requireNamespace("readxl", quietly = TRUE)) {
  install.packages("readxl")
}

library(readxl)


# ============================================================
# 2. IMPORT THE EXCEL DATASET
# ============================================================

energy <- read_excel(
"C:/Users/sunita.sharma/Downloads/Energy_Efficiency_Multivariate_Regression_data.xlsx"
)

energy <- as.data.frame(energy)

# Display the first six observations
head(energy)


# ============================================================
# 3. EXPLORE THE DATASET
# ============================================================
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# Examine variable types and structure
str(energy)

# Find the number of observations and variables
dim(energy)

# Display variable names
names(energy)

# Obtain summary statistics
summary(energy)


# ============================================================
# 4. CHECK FOR MISSING VALUES
# ============================================================

# Count missing values in each variable
colSums(is.na(energy))

# Remove observations with missing values in the variables
# used in the regression model

required_variables <- c(
  "Y1_Heating_Load",
  "Y2_Cooling_Load",
  "X1_Relative_Compactness",
  "X2_Surface_Area",
  "X3_Wall_Area",
  "X4_Roof_Area",
  "X5_Overall_Height",
  "X6_Orientation",
  "X7_Glazing_Area"
)

# Check that all required variables exist
missing_variables <- setdiff(
  required_variables,
  names(energy)
)

if (length(missing_variables) > 0) {
  stop(
    paste(
      "Missing variables:",
      paste(missing_variables, collapse = ", ")
    )
  )
}

energy <- energy[
  complete.cases(energy[, required_variables]),
]
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# ============================================================
# 5. CONVERT ORIENTATION INTO A CATEGORICAL VARIABLE
# ============================================================

# Orientation represents categories, not a continuous measurement.

energy$X6_Orientation <- as.factor(energy$X6_Orientation)

# Check the categories
levels(energy$X6_Orientation)

# Display the number of observations in each category
table(energy$X6_Orientation)


# ============================================================
# 6. DESCRIPTIVE STATISTICS OF THE RESPONSE VARIABLES
# ============================================================

# Summary of heating load
summary(energy$Y1_Heating_Load)

# Summary of cooling load
summary(energy$Y2_Cooling_Load)

# Mean heating load
mean(energy$Y1_Heating_Load)

# Mean cooling load
mean(energy$Y2_Cooling_Load)

# Standard deviation of heating load
sd(energy$Y1_Heating_Load)

# Standard deviation of cooling load
sd(energy$Y2_Cooling_Load)


# ============================================================
# 7. EXPLORE THE RELATIONSHIP BETWEEN THE TWO RESPONSES
# ============================================================

# Correlation between heating load and cooling load

cor(
  energy$Y1_Heating_Load,
  energy$Y2_Cooling_Load
)

# Scatterplot of the two response variables

plot(
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  energy$Y1_Heating_Load,
  energy$Y2_Cooling_Load,
  main = "Heating Load versus Cooling Load",
  xlab = "Heating Load",
  ylab = "Cooling Load",
  pch = 19
)


# ============================================================
# 8. FORMULATE THE MULTIVARIATE MULTIPLE REGRESSION MODEL
# ============================================================

# General model:
#
# Y1 = beta10 + beta11*X1 + beta12*X2 + ... + error1
#
# Y2 = beta20 + beta21*X1 + beta22*X2 + ... + error2
#
# The two responses are modelled together using cbind().
#
# The same predictor variables are used for both responses,
# but their estimated regression coefficients can differ.

model <- lm(
  cbind(Y1_Heating_Load, Y2_Cooling_Load) ~
    X1_Relative_Compactness +
    X2_Surface_Area +
    X3_Wall_Area +
    X4_Roof_Area +
    X5_Overall_Height +
    X6_Orientation +
    X7_Glazing_Area,
  data = energy
)


# ============================================================
# 9. DISPLAY THE REGRESSION RESULTS
# ============================================================

# Obtain the regression summaries for both responses

summary(model)

# The output provides coefficient estimates, standard errors,
# t-statistics, p-values, residual standard errors, and
# R-squared values for the individual response equations.
#
# Interpret each response separately:
# 1. Heating load
# 2. Cooling load


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# ============================================================
# 10. EXTRACT THE ESTIMATED REGRESSION COEFFICIENTS
# ============================================================

# Display the estimated coefficients for both responses

coefficients <- coef(model)

print(coefficients)


# ============================================================
# 11. WRITE THE FITTED REGRESSION EQUATIONS
# ============================================================

# A fitted regression equation uses estimated coefficients
# instead of unknown population parameters.
#
# For example:
#
# Fitted Y1 = estimated intercept
#             + estimated coefficient of X1 * X1
#             + estimated coefficient of X2 * X2 + ...
#
# The following function prints each fitted equation
# using the actual coefficient estimates from R.

print_equation <- function(coef_values, response_name) {

  # Round the coefficients for easier classroom interpretation
  coef_values <- round(coef_values, 4)

  # The first coefficient is the intercept
  intercept <- coef_values[1]

  # Check whether the intercept is missing
  if (is.na(intercept) || !is.finite(intercept)) {
    cat(
      "The intercept for",
      response_name,
      "cannot be estimated.\n"
    )
    return(invisible(NULL))
  }

  # The remaining coefficients correspond to predictors
  predictor_coefs <- coef_values[-1]

  predictor_names <- names(predictor_coefs)

  # Begin the equation with the intercept
  equation <- paste0(
    response_name, " = ", intercept
  )
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  # Add each predictor term with its estimated coefficient
  for (i in seq_along(predictor_coefs)) {

    b <- predictor_coefs[i]
    predictor <- predictor_names[i]

    # Skip missing or non-finite coefficients
    if (is.na(b) || !is.finite(b)) {
      next
    }

    if (b >= 0) {
      equation <- paste0(
        equation, " + ", abs(b), " * ", predictor
      )
    } else {
      equation <- paste0(
        equation, " - ", abs(b), " * ", predictor
      )
    }
  }

  cat(equation, "\n\n")
}

# Fitted equation for heating load

print_equation(
  coefficients[, "Y1_Heating_Load"],
  "Fitted Heating Load"
)

# Fitted equation for cooling load

print_equation(
  coefficients[, "Y2_Cooling_Load"],
  "Fitted Cooling Load"
)


# ============================================================
# 12. INTERPRET THE REGRESSION COEFFICIENTS
# ============================================================

# Display the coefficient table for heating load

summary(model)$`Response Y1_Heating_Load`

# Display the coefficient table for cooling load

summary(model)$`Response Y2_Cooling_Load`

# Interpretation:
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# A regression coefficient represents the estimated change
# in a response for a one-unit increase in a predictor,
# holding the other predictors constant.
#
# For categorical orientation, each coefficient represents
# the difference from the reference category, holding the
# other predictors constant.
#
# A positive coefficient indicates an increase in the
# predicted response, while a negative coefficient indicates
# a decrease, subject to the other predictors being fixed.


# ============================================================
# 13. OBTAIN THE FITTED VALUES
# ============================================================

# Calculate predicted values for both responses

fitted_values <- fitted(model)

# Display the first six fitted observations

head(fitted_values)

# Extract the fitted heating loads

fitted_heating <- fitted_values[, 1]

# Extract the fitted cooling loads

fitted_cooling <- fitted_values[, 2]


# ============================================================
# 14. OBTAIN THE RESIDUALS
# ============================================================

# Residual = Observed value - Fitted value

residual_values <- residuals(model)

# Display the first six residuals

head(residual_values)

# Residuals for heating load

residual_heating <- residual_values[, 1]

# Residuals for cooling load

residual_cooling <- residual_values[, 2]

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# ============================================================
# 15. COMPARE OBSERVED AND FITTED HEATING LOADS
# ============================================================

heating_results <- data.frame(
  Observed_Heating = energy$Y1_Heating_Load,
  Fitted_Heating = fitted_heating,
  Residual_Heating = residual_heating
)

head(heating_results)


# ============================================================
# 16. COMPARE OBSERVED AND FITTED COOLING LOADS
# ============================================================

cooling_results <- data.frame(
  Observed_Cooling = energy$Y2_Cooling_Load,
  Fitted_Cooling = fitted_cooling,
  Residual_Cooling = residual_cooling
)

head(cooling_results)


# ============================================================
# 17. PLOT OBSERVED VERSUS FITTED HEATING LOADS
# ============================================================

plot(
  energy$Y1_Heating_Load,
  fitted_heating,
  main = "Observed versus Fitted Heating Load",
  xlab = "Observed Heating Load",
  ylab = "Fitted Heating Load",
  pch = 19
)

abline(a = 0, b = 1, col = "red", lwd = 2)

# Points closer to the red line indicate closer agreement
# between observed and fitted values.


# ============================================================
# 18. PLOT OBSERVED VERSUS FITTED COOLING LOADS
# ============================================================

plot(
  energy$Y2_Cooling_Load,
  fitted_cooling,
  main = "Observed versus Fitted Cooling Load",
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  xlab = "Observed Cooling Load",
  ylab = "Fitted Cooling Load",
  pch = 19
)

abline(a = 0, b = 1, col = "red", lwd = 2)


# ============================================================
# 19. RESIDUALS VERSUS FITTED HEATING LOADS
# ============================================================

plot(
  fitted_heating,
  residual_heating,
  main = "Residuals versus Fitted Heating Load",
  xlab = "Fitted Heating Load",
  ylab = "Residuals",
  pch = 19
)

abline(h = 0, col = "red", lwd = 2)

# Look for random scatter around zero.
# A clear pattern may indicate a problem with model fit.


# ============================================================
# 20. RESIDUALS VERSUS FITTED COOLING LOADS
# ============================================================

plot(
  fitted_cooling,
  residual_cooling,
  main = "Residuals versus Fitted Cooling Load",
  xlab = "Fitted Cooling Load",
  ylab = "Residuals",
  pch = 19
)

abline(h = 0, col = "red", lwd = 2)


# ============================================================
# 21. HISTOGRAMS OF THE RESIDUALS
# ============================================================

# Heating load residuals

hist(
  residual_heating,
  main = "Histogram of Heating Load Residuals",
  xlab = "Residuals",
  col = "lightblue"
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)

# Cooling load residuals

hist(
  residual_cooling,
  main = "Histogram of Cooling Load Residuals",
  xlab = "Residuals",
  col = "lightgreen"
)

# Histograms can help assess whether the residuals
# are approximately symmetric and bell-shaped.


# ============================================================
# 22. CALCULATE THE MEAN SQUARED ERROR
# ============================================================

# MSE = Mean of squared residuals

MSE_heating <- mean(residual_heating^2)

MSE_cooling <- mean(residual_cooling^2)

cat("MSE for Heating Load =", MSE_heating, "\n")

cat("MSE for Cooling Load =", MSE_cooling, "\n")

# A smaller MSE indicates smaller average squared
# prediction errors on the data used to fit the model.


# ============================================================
# 23. CALCULATE R-SQUARED FOR EACH RESPONSE
# ============================================================

# R-squared measures the proportion of variation in
# a response explained by its regression model.

R2_heating <- 1 -
  sum(residual_heating^2) /
  sum((energy$Y1_Heating_Load -
         mean(energy$Y1_Heating_Load))^2)

R2_cooling <- 1 -
  sum(residual_cooling^2) /
  sum((energy$Y2_Cooling_Load -
         mean(energy$Y2_Cooling_Load))^2)

cat("R-squared for Heating Load =", R2_heating, "\n")

cat("R-squared for Cooling Load =", R2_cooling, "\n")

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# ============================================================
# 24. SAVE ALL OBSERVED AND FITTED VALUES
# ============================================================

results <- data.frame(
  Observed_Heating = energy$Y1_Heating_Load,
  Fitted_Heating = fitted_heating,
  Residual_Heating = residual_heating,
  Observed_Cooling = energy$Y2_Cooling_Load,
  Fitted_Cooling = fitted_cooling,
  Residual_Cooling = residual_cooling
)

# Display the first six rows

head(results)


# ============================================================
# 25. PRACTICE QUESTION
# ============================================================

# Question:
# Using the Energy Efficiency dataset, fit a multivariate
# multiple regression model with Heating Load and Cooling Load
# as the two response variables and the following predictors:
#
# X3_Wall_Area
# X4_Roof_Area
# X7_Glazing_Area
#
# Perform the following tasks:
#
# (a) Fit the multivariate multiple regression model.
#
# (b) Display the regression summaries for both response
#     variables.


```